Human-Readable Proof Outline for the Kasami APN Theorem (Odd Dimension)

RequestProject / Kasami APN

1 What is proved

The Lean theorem `kasami_is_apn` proves that the Kasami power map

$$f(x) = x^{d(k)}, \qquad d(k) = 2^{2k} - 2^k + 1,$$

is APN on a field F of characteristic 2 under the assumptions:

- $|F| = 2^n$,
- n odd,
- $1 < k < n$,
- k odd,
- $\gcd(k, n) = 1$.

Among these assumptions, the odd-dimension condition (n odd) becomes essential at Layer 5, where it is used to obtain the coprimality input needed to construct the linking exponent e'.

Also, at Layer 6, the Gold-map bijectivity is used to cancel a (2^k+1)-power, which is only valid under the coprimality assumption $\gcd(2^k + 1, 2^n - 1) = 1$.

2 APN meaning and reduction strategy

For a function $f : F \to F$, APN means that for every nonzero a and every collision

$$f(x + a) + f(x) = f(y + a) + f(y),$$

the only possible outputs are $y = x$ or $y = x + a$.

The proof does not attack this directly for all a. It first uses the normalization lemma `apn_of_normalized`: it is enough to prove the same uniqueness statement for the normalized equation with $a = 1$:

$$(x + 1)^d + x^d = (y + 1)^d + y^d \implies y = x \text{ or } y = x + 1.$$

So the whole theorem is reduced to controlling collisions of the normalized differential.

3 Layered proof architecture

The Lean file is organized in layers. The theorem-level dependency chain is:

$$\begin{array}{c}\texttt{kasami\_is\_apn}\\ \uparrow\\ \texttt{apn\_of\_normalized}\\ \uparrow\\ \texttt{kasami\_collision\_forces\_equal\_u}\\ \uparrow\\ \texttt{phi\_injective\_on\_units}\\ \uparrow\\ \texttt{exists\_linking\_exp}\end{array}$$

Additional inputs are the Artin–Schreier identity, the key Kasami polynomial identity, Gold-map bijectivity, and a quadratic-kernel fact in characteristic 2.

Layer 1: Artin–Schreier trace identity

`truncTrace_artin_schreier` proves

$$L_k(x^2 + x) = x^{2^k} + x.$$

This is the basic conversion between truncated trace expressions and explicit powers.

Layer 2: Key Kasami identity

`kasami_key_identity` proves

$$\big((x + 1)^d + x^d + 1\big)(x^2 + x)^{2^k} = (x^{2^k} + x)^{2^k+1}.$$

This is the bridge from differential terms to a form suited for the Φ-map analysis.

Layer 3: Gold coprimality and bijectivity

`gold_coprime` gives

$$\gcd(2^k + 1, 2^n - 1) = 1,$$

and `gold_pow_bijective` deduces that

$$y \mapsto y^{2^k+1}$$

is bijective on F.

This bijectivity is later used to cancel an outer $(2^k + 1)$-power.

Layer 4: Arithmetic transfer identity

`kasami_arith_identity` is a modular arithmetic identity tailored to the exponent transfer step. It aligns the Kasami exponent arithmetic with the Dempwolff–Mueller bijection hypothesis.

Layer 5: Existence of a linking exponent e'

`exists_linking_exp` constructs e' with two modular constraints:

$$e'(2^k+1) \equiv 2^n - 1 - 2^k \pmod{2^n-1},$$
$$(2^{n-1} - 2^{k-1} - 1)e' \equiv 2^{k-1} \pmod{2^n-1}.$$

Interpretation: e' is the arithmetic compatibility parameter that lets the map

$$\Phi(u) = \frac{L_k(u)^{2^k+1}}{u^{2^k}}$$

be rewritten as a composition of two bijective-style maps.

This is the first point in the proof where the hypothesis "n odd" is genuinely needed: it enters through the Gold coprimality step used in constructing e'.

Layer 6: Injectivity of Φ on nonzero elements

`phi_injective_on_units` proves that for $u, v \neq 0$, a collision in the Φ-equation forces $u = v$. The proof combines:

- the exponent link from Layer 5,
- Gold-map bijectivity to cancel (2^k+1) powers,
- Dempwolff–Mueller theorem `LxXk'_bijective` to conclude equality of inputs.

The Gold-map bijectivity is only valid under the coprimality assumption $\gcd(2^k+1, 2^n-1) = 1$.

Layer 7: Differential collision implies equal Artin–Schreier values

`kasami_collision_forces_equal_u` proves:

$$(x+1)^d + x^d = (y+1)^d + y^d \implies x^2 + x = y^2 + y.$$

It applies Layer 2 to rewrite the collision, then:

- handles zero cases directly,
- uses Layer 6 injectivity in the nonzero case,
- uses Layer 1 for trace-power conversion.

This is the key bridge from normalized differential collisions to a quadratic relation in $x + y$.

Layer 8: Characteristic-2 kernel and normalization lemma

There are two lemmas used in the final step:

- `sq_add_self_eq_zero_char2`:

$$u^2 + u = 0 \iff u \in \{0, 1\}.$$

- `apn_of_normalized`: proving the normalized differential uniqueness implies full APN.

Layer 9: Final theorem

In `kasami_is_apn`, the final argument is short:

1. Apply `apn_of_normalized`; reduce to the $a = 1$ collision equation.
2. Use Layer 7 to get $x^2 + x = y^2 + y$.
3. Rearrange to
$$(x + y)^2 + (x + y) = 0.$$
4. Use Layer 8 kernel lemma to conclude $x + y = 0$ or $x + y = 1$.
5. Therefore $y = x$ or $y = x + 1$, exactly the APN condition.

4 One-paragraph summary

The proof is modular. It first reduces APN to a normalized differential equation, then converts that equation into a trace/exponent form via the Kasami identity, proves injectivity of the associated Φ-map on nonzero elements using a specially constructed linking exponent and known bijectivity theorems, and finally translates the resulting equality into the two-point alternative $y \in \{x, x + 1\}$ through a characteristic-2 kernel identity. This closes the APN theorem for Kasami exponents in odd dimension under the standard coprimality assumptions.